Misstatement Detection Lag and Prediction Evaluation

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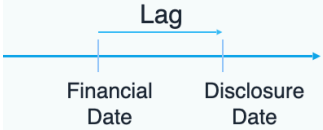
Summary

PROBLEM

Misstatement Detection Lag

Look-ahead Bias



Model setup

Conventional Approach	Continuously Updating Models	Two-year Gap
look-ahead bias	noisy zeros	non-update

Continuously updating models account for detection lags.

PERFORMANCE

Conventional Approach > **Continuously Updating Models** > **Two-year Gap**

Research Question

- How do misstatement detection lags impact the performance evaluation of misstatement prediction models?
  - Misstatement detection lags lead to "look-ahead bias" in model predictions and substantially inflate prediction performance.

Motivation

- Effectively detecting and preventing accounting misstatements is crucial in preserving investors' confidence and capital market efficiency.
  - Much of this prior work has focused on uncovering firm characteristics and incentives that lead to misreporting
  - a growing body of literature has shifted attention toward predictive modeling(ML)
- considerable attention has been given to performance inflation arising from serial misstatements (i.e., misstatements that span multiple consecutive years)
  - some studies propose leaving a temporal gap between training and test periods (e.g., a two-year buffer)

Motivation

- two critical issues remain underexplored in the literature
  - the look-ahead bias arising from misstatement detection lags
  - the time-varying nature of the misstatement data-generating process
- Look-ahead bias giving models an unfair advantage by using future information not available in real time
- This paper design an approach that accounts for detection lags and continuously updates models to adapt to the changing datagenerating process.

Contribution

1. literature on how to deal with detection lag
   - Prior: leaving a temporal gap between training and test periods; isolates a group of firms from the training sample(Bao et al., 2020; Bertomeu, 2020).
   - Extension: offering a framework for evaluating misstatement prediction models in a dynamic regulatory environment.
2. literature on the practical utility and economic implications of model predictions
   - Prior: Developing other prediction performance measures, such as return-based or cost-based evaluation measures (Beneish and Vorst 2022).
   - Extension: implement a trading strategy and shows that high-misstatement-risk generate significant and negative portfolio alphas.

Hypothesis

1. the conventional approach outperform the continuously updating approach
   - look-ahead bias
2. the two-year training-test gap underperform the continuously updating approach
   - data-generating process is time-varying/dynamic

Measuring Misstatement Detection Lag

- detection lag: the material misstatement disclosure date - the financial statement disclosure date
  - Misstatements date: the first 8-K form with Item 4.02 included to disclose the material misrepresentation of accounting numbers.
  - For irregularities, use the date of the first filing containing text information that meets one of the three criteria set up by Hennes et al. (2008)

Panel B: Distribution of the Detection Lags (Measured in Years)

Min	10th Percentile	Q1	Median	Q3	90th Percentile	Max
0.016	0.427	0.929	1.762	2.947	3.956	6.608

Prediction Models and Performance Criteria

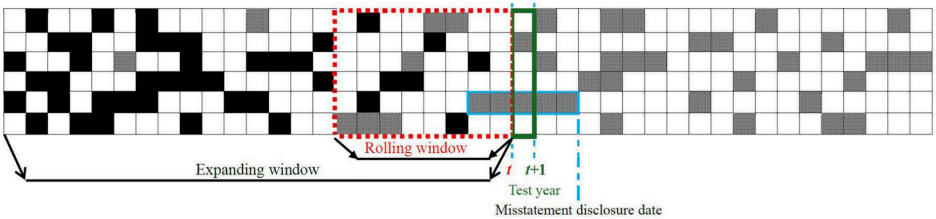
- Prediction Models
  - logistic regression(Dechow et al. 2011): model interpretability and prediction stability due to its pre-specified function form
  - RF(Breiman, 2001) and GBT(Friedman, 2001): the flexibility of tuning multiple hyperparameters to achieve optimal predictions for a specific sample
  - BRF: adopts an under-sampling scheme to ensure each subsample contains balanced misstating and nonmisstating observations
- Performance Criteria
  - ROC-AUC; PR-AUC; NDCG@k;

Continuously Updating Approach

- Assuming the training window ends at the end of year t, prediction models are only trained on the misstatements discovered within the training window by the end of year t.
- This process is repeated for each prediction year from 2004 to 2015.
- noisy zero: undetected misstatements incorrectly labeled as zeros.

centering

The Timeline of Training and Prediction Process in the Continuously Updating Approach



Conventional Approach

- Assuming the training window ends at the end of year t, all misstatement cases are known to the modeler at the time of model building, regardless of when they were discovered.
- This process is repeated for each prediction year from 2004 to 2015.
- Expanding window and three-year rolling windows with the continuously updating approach.
- look ahead bias: unfair inflate prediction performance.

Data and Implementation

- Sample
  - publicly traded U.S. firms listed on the three major exchanges (i.e., NYSE, NASDAQ, and AMEX) from 2001 to 2014.
  - 44,206 firm-year observations with 2,402 material restatements
- Data
  - restatement: 19,616 unique restatement files from the restatement data base in Audit Analytics (AA) in November 2021
  - predictors: 14 financial ratios and 28 raw accounting numbers(Bao et al. (2020)); market variables and audit variables(Bertomeu et al. (2021))

The conventional approach outperform the continuously updating

- due to the look-ahead bias

Approach	Model	Expanding Windows ROC-AUC	Expanding Windows PR-AUC	Expanding Windows NDCG@k	Rolling Windows ROC-AUC	Rolling Windows PR-AUC	Rolling Windows NDCG@k
Conventional	(1) Logit	0.701	0.112	0.095	0.718	0.121	0.102
	(2) RF	0.778	0.192	0.245	0.780	0.194	0.271
	(3) BRF	0.708	0.124	0.123	0.710	0.126	0.112
	(4) GBT	0.763	0.187	0.234	0.775	0.192	0.256
Continuously updating	(5) Logit	0.687	0.101	0.093	0.707	0.114	0.090
	(6) RF	0.748	0.162	0.195	0.751	0.165	0.205
	(7) BRF	0.689	0.102	0.104	0.692	0.107	0.098
	(8) GBT	0.745	0.160	0.179	0.750	0.162	0.192
		Performance Difference%					
Comparison	(9) Logit	7.49%	10.89%	2.15%	5.31%	6.14%	13.33%
	(10) RF	12.10%	18.52%	25.64%	11.55%	17.58%	32.20%
	(11) BRF	10.05%	21.57%	18.27%	9.38%	17.76%	14.29%
	(12) GBT	7.35%	16.88%	30.73%	10.00%	18.52%	33.33%

The two-year training-test gap underperforms the continuously updating approach

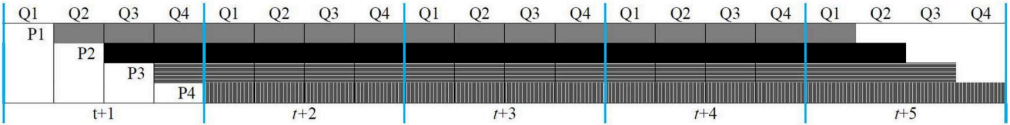
- due to data-generating process

Approach	Model	Expanding Windows ROC-AUC	Expanding Windows PR-AUC	Expanding Windows NDCG@k	Rolling Windows ROC-AUC	Rolling Windows PR-AUC	Rolling Windows NDCG@k
Two-year gap	(1) Logit	0.676	0.094	0.083	0.637	0.061	0.066
	(2) RF	0.677	0.092	0.122	0.639	0.055	0.090
	(3) BRF	0.645	0.076	0.083	0.616	0.051	0.061
	(4) GBT	0.696	0.108	0.124	0.644	0.059	0.093
Continuously updating	(5) Logit	0.687	0.101	0.093	0.707	0.114	0.090
	(6) RF	0.748	0.162	0.195	0.751	0.165	0.205
	(7) BRF	0.689	0.102	0.104	0.692	0.107	0.098
	(8) GBT	0.745	0.160	0.179	0.750	0.162	0.192
		Performance Difference%					
Comparison	(9) Logit	−5.88%	−6.93%	−10.75%	−33.82%	−46.49%	−26.67%
	(10) RF	−28.63%	−43.21%	−37.44%	−44.62%	−66.67%	−56.10%
	(11) BRF	−23.28%	−25.49%	−20.19%	−39.58%	−52.34%	−37.76%
	(12) GBT	−20.00%	−32.50%	−30.73%	−42.40%	−63.58%	−51.56%

Trading strategy

- Evaluate the practical utility of model predictions under the continuously updating approach.
- High-Misstatement-Risk Portfolio: Formed *quarterly* by selecting firms in the top 5% of predicted misstatement probabilities. Held for **four years (16 quarters)**.
- High-Risk & High-Detection-Lag Portfolio: Firms with top 5% misstatement risk and predicted detection lags above the sample median.

The Timeline of the Trading Strategy for the Continuously Updating Approach



High-misstatement-risk generate negative portfolio alphas

$$P_RET_m = \alpha + \beta_1 MKT\text{-}RF + \beta_2 SMB + \beta_3 HML + \beta_4 RMW + \beta_5 CMA + \beta_6 UMD + \varepsilon_m$$

Model	*alpha*	*MKT_RF*	*SMB*	*HML*	*RMW*	*CMA*	*UMD*
High-misstatement-risk Portfolio (1)							
Logit	−0.001	1.070	0.822	−0.362	0.341	−0.330	−0.370
	(−1.13)	(23.38)	(10.84)	(−3.84)	(2.23)	(−8.04)	(−3.20)
RF	−0.004	1.003	0.631	−0.362	0.438	−0.329	−0.257
	(−2.08)	(19.38)	(7.16)	(−3.66)	(2.50)	(−6.83)	(−2.04)
BRF	0.000	0.932	0.433	−0.372	0.138	−0.359	−0.641
	(0.17)	(18.42)	(4.81)	(−3.44)	(0.83)	(−8.05)	(−4.79)
GBT	−0.003	0.918	0.705	−0.520	0.409	−0.327	−0.424
	(−1.99)	(19.84)	(9.56)	(−5.88)	(3.00)	(−7.45)	(−3.70)
High-misstatement-risk, high-detection-lag Portfolio (2)							
Logit	−0.005	1.149	0.791	−0.346	0.536	−0.324	−0.305
	(−2.44)	(20.16)	(8.93)	(−3.04)	(2.95)	(−6.31)	(−2.02)
RF	−0.006	1.037	0.626	−0.466	0.373	−0.352	−0.255
	(−2.53)	(17.06)	(6.03)	(−4.18)	(2.10)	(−6.57)	(−1.63)
BRF	−0.004	0.903	0.406	−0.377	0.175	−0.364	−0.608
	(−1.97)	(16.01)	(3.85)	(−3.12)	(0.96)	(−7.23)	(−4.05)
GBT	−0.005	1.002	0.747	−0.524	0.389	−0.284	−0.208
	(−2.46)	(16.07)	(7.34)	(−4.04)	(1.95)	(−5.22)	(−1.23)

Ideas

- 本文认为由于财务错报被发现存在滞后问题，所以从模型设计的角度处理；
- 直接从数据角度处理财务错报滞后问题，从更加及时发布的企业相关新闻文本及时挖掘企业财务错报概率